Optimization: Critical Points & Lagrange Multipliers

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Finding Critical Points**

Find all critical points of the function $f(x, y) = x^3 + y^3 - 3xy$.

1. Set the first partial derivatives equal to zero: $f_x = 3x^2 - 3y = 0 \Rightarrow y = x^2$. $f_y = 3y^2 - 3x = 0 \Rightarrow x = y^2$. 2. Substitute $y = x^2$ into the second equation: $x = (x^2)^2 = x^4 \Rightarrow x^4 - x = 0 \Rightarrow x(x^3 - 1) = 0$. Real solutions are $x = 0$ and $x = 1$. 3. Determine corresponding y-values:If $x = 0$, $y = 0^2 = 0 \Rightarrow (0, 0)$.If $x = 1$, $y = 1^2 = 1 \Rightarrow (1, 1)$.Thus, the two critical points are $(0, 0)$ and $(1, 1)$.

2 **[Foundational]** • **Target: The Second Partials Test (Hessian Discriminant)**

Consider the surface S which is the portion of the plane $z = 2x + 3y + 1$ lying above the rectangular region $R = [0, 1] \times [0, 2]$ in the xy-plane. Calculate the surface area of S.

The surface area of a surface $z = g(x, y)$ above a region R is given by the integral $\iint_R \sqrt{1 + (g_x)^2 + (g_y)^2}\, dA$. Here, $g(x, y) = 2x + 3y + 1$. The partial derivatives are $g_x = 2$ and $g_y = 3$. Therefore, $\sqrt{1 + (g_x)^2 + (g_y)^2} = \sqrt{1 + 2^2 + 3^2} = \sqrt{1 + 4 + 9} = \sqrt{14}$. The integral becomes $\iint_R \sqrt{14}\, dA$, where $R = [0, 1] \times [0, 2]$. This evaluates to $\sqrt{14} \times \text{Area}(R)$. The area of the rectangle R is $(1 - 0) \times (2 - 0) = 2$. So, the sur.

3 **[Foundational]** • **Target: Classifying Extrema and Inconclusive Cases**

Compute the scalar line integral with respect to arc length $\int_C (x + 2y + 3z)\, ds$, where C is the straight line segment from the origin $(0, 0, 0)$ to $(1, 2, 2)$.

1. Parametrize the line segment: $\mathbf{r}(t) = t\langle 1, 2, 2\rangle = \langle t, 2t, 2t\rangle, \quad 0 \le t \le 1$. 2. Compute differential arc length ds: $\mathbf{r}'(t) = \langle 1, 2, 2\rangle \Rightarrow ||\mathbf{r}'(t)|| = \sqrt{1^2 + 2^2 + 2^2} = \sqrt{9} = 3 \Rightarrow ds = 3\, dt$. 3. Evaluate integrand along C: $x + 2y + 3z = t + 2(2t) + 3(2t) = t + 4t + 6t = 11t$. 4. Integrate.

4 **[Foundational]** • **Target: Lagrange Multipliers for Constrained Optimization**

Consider the function $f(x, y) = x^3 - 3xy + y^2$. Find the value of $f_{xx}(1, 1)$.

To find $f_{xx}(1, 1)$, we first need to compute the first partial derivative of f with respect to x, which is f_x. Then, we compute the second partial derivative of f with respect to x, which is f_{xx}. Finally, we evaluate f_{xx} at the point $(1, 1)$. First partial derivative with respect to x: $f_x = \frac{\partial}{\partial x}(x^3 - 3xy + y^2) = 3x^2 - 3y$. Second partial derivative with respect to x: $f_{xx} = \frac{\partial}{\partial x}(3x^2 - 3y) = 6x$. Now, evaluate f_{xx} at $(1, 1)$.

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5 [Foundational] • Target: Finding Critical Points

Let the scalar field be given by $f(x, y) = 3x^2 - 5y$. Compute the value of $f_x(2, -3)$.

To find $f_x(2, -3)$, we first compute the partial derivative of $f(x, y)$ with respect to x. Treating y as a constant, we get: $f_x(x, y) = \frac{\partial}{\partial x}(3x^2 - 5y) = 6x$. Now, we evaluate f_x at the point $(2, -3)$: $f_x(2, -3) = 6(2) = 12$.

6 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

Evaluate $\int_C \mathbf{F} \cdot d\mathbf{r}$ for $\mathbf{F} = \langle x^2, xy \rangle$ along the segment from $(3, 3)$ to $(1, 0)$.

Parametrize $\mathbf{r}(t) = \langle 3 - 2t, 3 - 3t \rangle,\ 0 \le t \le 1$. Then $\int_0^1 \mathbf{F}(\mathbf{r}(t)) \cdot \mathbf{r}'(t)\, dt = \int_0^1 (-26t^2 + 69t - 45)\, dt = -\frac{115}{6}$.

7 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

Find the volume of the solid inside the sphere $\rho = 4$ and above the cone $\phi = \frac{\pi}{4}$.

$V = \int_0^{2\pi} \int_0^{\pi/4} \int_0^4 \rho^2 \sin\phi \, d\rho \, d\phi \, d\theta = 2\pi\left(\frac{64}{3} - \frac{32\sqrt{2}}{3}\right)$. Final Answer: $2\pi\left(\frac{64}{3} - \frac{32\sqrt{2}}{3}\right)$

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

Compute $\iint_D 9(x^2 + y^2)\, dA$ where D is the disk $x^2 + y^2 \le 49$.

In polar coordinates, D is $0 \le r \le 7$, $0 \le \theta \le 2\pi$: $\iint_D 9(x^2 + y^2)\, dA = 9\int_0^{2\pi}\int_0^7 r^3\, dr\, d\theta = 9\int_0^{2\pi} \frac{7^4}{4} d\theta = 21609\pi/2$.

9 [Intermediate] • Target: Finding Critical Points

For $f(x, y) = 4x^2y^2 + 3xy$, find the mixed partial $f_{xy}(1, -2)$.

$f_x = 8xy^2 + 3y$, so $f_{xy} = \frac{\partial}{\partial y} f_x = 16xy + 3$. At $(1, -2)$: $f_{xy} = -29$ (matches f_{yx}, as Clairaut's theorem guarantees since f is a polynomial).

10 [Intermediate] • Target: The Second Partials Test (Hessian Discriminant)

Use Green's theorem to evaluate $\oint_C -y^3\, dx + x^3\, dy$, where C is the triangle with vertices $(0, 0)$, $(3, 0)$, $(0, 3)$, oriented counterclockwise.

$\oint_C P\, dx + Q\, dy = \iint_D (Q_x - P_y)\, dA$ with $Q_x - P_y = 3x^2 + 3y^2$. $\int_0^3 \int_0^{3-x} (3x^2 + 3y^2)\, dy\, dx = \frac{81}{2}$.

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11 [Intermediate] • Target: Classifying Extrema and Inconclusive Cases

Consider the function $f(x, y) = x^2 + y^2 - 4x - 6y + 13$. Find the coordinates of the critical point for this function.

Final Answer: (2, 3). To find the critical point(s) of a function of two variables, we need to find where the gradient is zero. The gradient is given by $ablaf = \left\langle \frac{\partial f}{\partial x}, \frac{\partial f}{\partial y} \right\rangle$. First, we compute the partial derivative with respect to x: $\frac{\partial f}{\partial x} = \frac{\partial}{\partial x}(x^2 + y^2 - 4x - 6y + 13) = 2x - 4$. Next, we compute the partial derivative with respect to y.

12 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

Consider the function $f(x, y) = x^3 + y^3 - 3axy$, where a is a positive constant. It is known that $f(x, y)$ has a critical point at (x_0, y_0) where $x_0 = 2$ and $y_0 = 2$. Find the value of a.

Final Answer: 2. To find the critical points of the function $f(x, y) = x^3 + y^3 - 3axy$, we first compute the first partial derivatives and set them to zero: $f_x = 3x^2 - 3ay$ $f_y = 3y^2 - 3ax$ Setting these to zero, we get the system of equations: 1) $3x^2 - 3ay = 0$ => $x^2 = ay$ 2) $3y^2 - 3ax = 0$ => $y^2 = ax$ We are given that $(x_0, y_0) = (2, 2)$ is a critical point. Substituting these values into the equations: From equation 1: $2^2 = a(2)$ => $4 = 2a$ => $a = 2$. From equation 2: $2^2 = a(2)$ =>.

13 [Intermediate] • Target: Finding Critical Points

Consider the function $f(x, y) = x^3 - 3xy + y^2$. Find the value of $f_{xx}(1, 1)$.

Final Answer: 6. To find $f_{xx}(1, 1)$, we first need to compute the first partial derivative of f with respect to x, which is f_x. Then, we compute the second partial derivative of f with respect to x, which is f_{xx}. Finally, we evaluate f_{xx} at the point $(1, 1)$. First partial derivative with respect to x: $f_x = \frac{\partial}{\partial x}(x^3 - 3xy + y^2) = 3x^2 - 3y$. Second partial derivative with respect to x: $f_{xx} = \frac{\partial}{\partial x}(3x^2 - 3y) = 6x$. Now, evaluate.

14 [Intermediate] • Target: The Second Partials Test (Hessian Discriminant)

Consider the function $f(x, y) = x^2 + y^2$. Find the value of $f_{xx}(2, 3)$.

Final Answer: 2. First, we find the first partial derivative of $f(x, y)$ with respect to x: $f_x(x, y) = \frac{\partial}{\partial x}(x^2 + y^2) = 2x$. Next, we find the second partial derivative of $f(x, y)$ with respect to x (which is f_{xx}): $f_{xx}(x, y) = \frac{\partial}{\partial x}(f_x(x, y)) = \frac{\partial}{\partial x}(2x) = 2$. Finally, we evaluate $f_{xx}(x, y)$ at the point $(2, 3)$. Since $f_{xx}(x, y)$ is a constant, its value at any point is 2. Therefore, $f_{xx}(2, 3) = 2$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

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15 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -24. $f_{xx} = 2$, $f_{yy} = 6$, $f_{xy} = -6$, so $D = 2(6) - (-6)^2 = -24$.

16 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 3x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -24. $f_{xx} = 6$, $f_{yy} = 2$, $f_{xy} = -6$, so $D = 6(2) - (-6)^2 = -24$.

17 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -32. $f_{xx} = 2$, $f_{yy} = 2$, $f_{xy} = -6$, so $D = 2(2) - (-6)^2 = -32$.

18 [Challenging] • Target: The Second Partials Test (Hessian Discriminant) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 2x^2 - 6xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -20. $f_{xx} = 4$, $f_{yy} = 4$, $f_{xy} = -6$, so $D = 4(4) - (-6)^2 = -20$.

19 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 6xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -20. $f_{xx} = 8$, $f_{yy} = 2$, $f_{xy} = -6$, so $D = 8(2) - (-6)^2 = -20$.

20 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 3x^2 - 5xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

Final Answer: -13. $f_{xx} = 6$, $f_{yy} = 2$, $f_{xy} = -5$, so $D = 6(2) - (-5)^2 = -13$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Finding Critical Points

What condition defines a critical point (x_0, y_0) of a differentiable function $f(x, y)$?

a. $\nabla f(x_0, y_0) = \mathbf{0}$**, meaning both** $f_x(x_0, y_0) = 0$ **and** $f_y(x_0, y_0) = 0$ ✓

b. $f_{xx}(x_0, y_0) = 0$

c. $f(x_0, y_0) = 0$

d. $f_x(x_0, y_0) = f_y(x_0, y_0)$

Critical points occur where the gradient vector is zero (both first-order partials vanish simultaneously) or where a partial derivative fails to exist.

22 [Foundational] • Target: Finding Critical Points

Find the critical points of $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. $(2, -3)$ ✓

b. $(-2, 3)$

c. $(4, -6)$

d. $(0, 0)$

$f_x = 2x - 4 = 0 \Rightarrow x = 2$. $f_y = 2y + 6 = 0 \Rightarrow y = -3$. The only critical point is $(2, -3)$.

23 [Foundational] • Target: Finding Critical Points

Find all critical points of $g(x, y) = x^3 - 3x + y^2$.

a. $(1, 0)$ **and** $(-1, 0)$ ✓

b. $(1, 0)$ only

c. $(0, 0)$

d. $(3, 0)$

$g_x = 3x^2 - 3 = 0 \Rightarrow x^2 = 1 \Rightarrow x = \pm 1$. $g_y = 2y = 0 \Rightarrow y = 0$. The critical points are $(1, 0)$ and $(-1, 0)$.

24 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

What is the definition of the Hessian discriminant D in the Second Partials Test?

a. $D = f_{xx}f_{yy} - (f_{xy})^2$ ✓

b. $D = f_{xx} + f_{yy}$

c. $D = (f_{xy})^2 - f_{xx}f_{yy}$

d. $D = f_{xx}f_{yy} - f_{xy}$

The discriminant D is the determinant of the Hessian matrix $[[f_{xx}, f_{xy}], [f_{yx}, f_{yy}]]$, which is $D = f_{xx}f_{yy} - (f_{xy})^2$.

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25 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

If at a critical point $D > 0$ and $f_{xx} > 0$, what does the test conclude?

a. f **has a local minimum at the point ✓**

b. f has a local maximum at the point

c. The point is a saddle point

d. The test is inconclusive

When $D > 0$, the surface is locally bowl-shaped; if $f_{xx} > 0$, it curves upward in all directions, establishing a local minimum.

26 [Foundational] • Target: The Second Partials Test (Hessian Discriminant)

If at a critical point $D < 0$, what does the test conclude?

a. f **has a saddle point ✓**

b. f has a local maximum

c. f has a local minimum

d. The test is inconclusive

When $D < 0$, the surface curves upward in one direction and downward in another, producing a saddle point.

27 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

Classify the critical point $(0, 0)$ for $f(x, y) = x^2 - y^2$.

a. **Saddle point ✓**

b. Local minimum

c. Local maximum

d. Inconclusive

$f_x = 2x, f_y = -2y \Rightarrow (0, 0)$. $f_{xx} = 2, f_{yy} = -2, f_{xy} = 0$. $D = (2)(-2) - 0^2 = -4 < 0$. Therefore $(0, 0)$ is a saddle point.

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28 [Foundational] • Target: Classifying Extrema and Inconclusive Cases

What does the Second Partials Test conclude when $D = 0$?

a. The test is inconclusive; higher-order analysis or algebraic examination is needed ✓

b. The point is definitely a saddle point

c. The point is an absolute extremum

d. The function is not differentiable

When $D = 0$, the test yields no information; the critical point could be a local maximum, local minimum, or saddle point.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Classifying Extrema and Inconclusive Cases

Classify $(2, -3)$ for $f(x, y) = x^2 + y^2 - 4x + 6y + 10$.

a. Local minimum ✓

b. Local maximum

c. Saddle point

d. Inconclusive

$f_{xx} = 2, f_{yy} = 2, f_{xy} = 0$. $D = (2)(2) - 0 = 4 > 0$. Since $f_{xx} = 2 > 0$, it is a local minimum.

30 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

What vector equation governs constrained optimization of $f(x, y)$ subject to constraint $g(x, y) = k$?

a. $\nabla f = \lambda \nabla g$ (and $g(x, y) = k$) ✓

b. $\nabla f \cdot \nabla g = 0$

c. $\nabla f + \nabla g = \mathbf{0}$

d. $\nabla f = \mathbf{0}$

At a constrained extremum, the level curve of f is tangent to the constraint curve $g = k$, meaning their gradient vectors must be parallel: $\nabla f = \lambda \nabla g$.

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31 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

Maximize $f(x, y) = xy$ subject to the constraint $x + y = 10$ for $x, y > 0$.

a. Maximum is 25 at $(5, 5)$ ✓

b. Maximum is 24 at $(4, 6)$

c. Maximum is 21 at $(3, 7)$

d. Maximum is 16 at $(2, 8)$

$\nabla f = [y, x]$ and $\nabla g = [1, 1]$. $[y, x] = \lambda[1, 1] \Rightarrow y = \lambda$ and $x = \lambda \Rightarrow x = y$. Using $x + y = 10 \Rightarrow 2x = 10 \Rightarrow x = 5, y = 5$. Maximum $f(5, 5) = 25$.

32 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

What does the Lagrange multiplier λ represent physically/economically in constrained optimization?

a. The sensitivity / shadow price: the rate of change of the optimal value with respect to the constraint constant k ($\frac{df^*}{dk}$) ✓

b. The curvature of the constraint surface

c. The distance from the origin to the optimal point

d. The second derivative along the constraint

The multiplier $\lambda = \frac{\partial f^*}{\partial k}$ represents the marginal change in the optimized objective value per unit increase in the constraint resource k.

33 [Intermediate] • Target: Finding Critical Points

For $f(x, y) = 2x^2 - 6xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 24

b. -12 ✓

c. -24

d. 12

$f_{xx} = 4$, $f_{yy} = 6$, $f_{xy} = -6$, so $D = 4(6) - (-6)^2 = -12$.

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34 [Intermediate] • Target: The Second Partials Test (Hessian Discriminant)

For $f(x, y) = 4x^2 - 2xy + y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. **12** ✓

b. 16

c. 8

d. -12

$f_{xx} = 8$, $f_{yy} = 2$, $f_{xy} = -2$, so $D = 8(2) - (-2)^2 = 12$.

35 [Intermediate] • Target: Classifying Extrema and Inconclusive Cases

For $f(x, y) = 3x^2 - xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 36

b. 33

c. **35** ✓

d. -35

$f_{xx} = 6$, $f_{yy} = 6$, $f_{xy} = -1$, so $D = 6(6) - (-1)^2 = 35$.

36 [Intermediate] • Target: Lagrange Multipliers for Constrained Optimization

For $f(x, y) = 2x^2 - xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

b. **31** ✓

c. 29

d. -31

$f_{xx} = 4$, $f_{yy} = 8$, $f_{xy} = -1$, so $D = 4(8) - (-1)^2 = 31$.

37 [Intermediate] • Target: Finding Critical Points

For $f(x, y) = 2x^2 - xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 16

b. 13

c. **15** ✓

d. -15

$f_{xx} = 4$, $f_{yy} = 4$, $f_{xy} = -1$, so $D = 4(4) - (-1)^2 = 15$.

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Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: The Second Partials Test (Hessian Discriminant) • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 3x^2 - 3xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 36

b. **27** ✓

c. 21

d. -27

$f_{xx} = 6$, $f_{yy} = 6$, $f_{xy} = -3$, so $D = 6(6) - (-3)^2 = 27$.

39 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 3x^2 - 2xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 24

b. 16

c. -20

d. **20** ✓

$f_{xx} = 6$, $f_{yy} = 4$, $f_{xy} = -2$, so $D = 6(4) - (-2)^2 = 20$.

40 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 2x^2 - 2xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

b. 24

c. **28** ✓

d. -28

$f_{xx} = 4$, $f_{yy} = 8$, $f_{xy} = -2$, so $D = 4(8) - (-2)^2 = 28$.

41 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = x^2 - 4xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 8

b. **-8** ✓

c. -16

d. -7

$f_{xx} = 2$, $f_{yy} = 4$, $f_{xy} = -4$, so $D = 2(4) - (-4)^2 = -8$.

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42 [Challenging] • Target: The Second Partials Test (Hessian Discriminant) • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 2xy + 2y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 32

b. **28** ✓

c. 24

d. -28

$f_{xx} = 8$, $f_{yy} = 4$, $f_{xy} = -2$, so $D = 8(4) - (-2)^2 = 28$.

43 [Challenging] • Target: Classifying Extrema and Inconclusive Cases • Cross-Lesson: 1.01 Functions of Several Variables & Partial Derivatives

For $f(x, y) = 4x^2 - 5xy + 3y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 48

b. 13

c. -23

d. **23** ✓

$f_{xx} = 8$, $f_{yy} = 6$, $f_{xy} = -5$, so $D = 8(6) - (-5)^2 = 23$.

44 [Challenging] • Target: Lagrange Multipliers for Constrained Optimization • Cross-Lesson: 2.01 The Chain Rule & Directional Derivatives

For $f(x, y) = 4x^2 - 2xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 64

b. 56

c. **60** ✓

d. -60

$f_{xx} = 8$, $f_{yy} = 8$, $f_{xy} = -2$, so $D = 8(8) - (-2)^2 = 60$.

45 [Challenging] • Target: Finding Critical Points • Cross-Lesson: 4.01 Double Integrals

For $f(x, y) = 4x^2 - 4xy + 4y^2$, compute the second-derivative-test discriminant $D = f_{xx}f_{yy} - f_{xy}^2$ (the same at every point, since all second partials here are constant).

a. 64

b. 40

c. -48

d. **48** ✓

$f_{xx} = 8$, $f_{yy} = 8$, $f_{xy} = -4$, so $D = 8(8) - (-4)^2 = 48$.

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